i: 0 ⟶ n-1

Binary Search for (tar - arr[i]) in arr[i+1 ... n-1].

TC = O(n log n).

AS = O(1)

i: 0 ⟶ n-1

if (tar - arr[i]) in Map:

return true

Map[arr[i]] = true.

TC = O(n)

AS = O(n).

[1, 2, 5, 6, 10], target: 8

2-ptr. →

TC = O(n)

AS = O(1)

[1, 2, 5, 6, 10]. tar = 9.

---

i: 0 ⟶ n-1

j: i+1 ⟶ n-1

k: j+1 ⟶ n-1

TC = O(n³)

AS = O(1).

Using 2-ptr:

Sort → O(n log n).

2-ptr → TC → O(n²)

AS → O(1).

Overall → O(n²) time.

i: 0 ⟶ n-1

2-ptr on arr[i+1 ... n-1]

to get pair with sum = target - arr[i].

TODO.

	Brute force	Optimal
k-sum problem	$O(n^k)$	$O(n^{k-1})$.